

# Název přednášky

Biostatistika - \_L-template

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Moodle

# Výsledky učení

Po absolvování této přednášky budete schopni:

- TODO: výsledek učení 1
- TODO: výsledek učení 2
- TODO: výsledek učení 3

# Obsah přednášky

# Snímek 1

TODO: obsah snímku

```
1 # TODO: ukázka kódu v R
```

# Shrnutí

- TODO: klíčové body přednášky

# Slide Creator's Guide

Slides in this section are intended for lecture authors, not students. Delete this section and the `# Tools Overview` section before using as an actual lecture.

# Golden Rules

# Rule 1 — Data story first

## WRONG

### Pearson correlation

Pearson's correlation coefficient measures the strength of the linear relationship between two variables...  
[formula]

## RIGHT

### How do we turn the shape of a cloud into a number?

[scatter plot the students have seen for several slides already]  
*What properties would such a number need to have?*

→ **Correlation coefficient** the name arrives as the answer

A statistical term never opens a concept as a heading. Data first → observation → question → only then the name.



# Rule 2 — One idea per slide

**Every slide communicates one thing.**

If you need to say two things, use two slides.

If you need to say three things, add an interaction between them.

**One message. One slide. Contribute to understanding — not to overload.**

Test: cover the heading — is the main idea of the slide still clear from the visual or a single sentence?

# Rule 3 — Interaction every 10–15 minutes

## SIX INTERACTION TYPES

| Type                | When to use                                                    |
|---------------------|----------------------------------------------------------------|
| Prediction question | Before revealing a result, graph, or test output               |
| Conceptual MCQ      | To surface a common misconception; ideally with a polling tool |
| Think–Pair–Share    | After a new concept; 30 s thinking + 1–2 min pair discussion   |
| Silent write        | Before discussion or reveal; each student writes one sentence  |
| Interpretation task | After showing a graph, model output, or table                  |

# Rule 4 — Always frame biologically

Every statistical method must be grounded in a concrete biological question.

**Four steps of the biological frame:**

1. **Biological question** — “we want to find out whether...”
2. **Data** — description, type, source
3. **Statistical method** — what and why
4. **Interpretation in biological context** — what the result says about the world, not about numbers

Never start with “today we’ll learn the t-test”. Always start with “we want to know whether...”

# Rule 5 — Alternate visual patterns (3-of-5 rule)

**At most 3 of 5 consecutive slides may use a named component as the primary visual element.**

A plain slide — a bold question on an empty background, a full-width graph, a single prominent number — carries more weight precisely because the surrounding slides are more structured.

**3 of 5 — the law of slide visual contrast** · review the block before finalising

Components: `.card-*`, `.panel-*`, `.blockquote`, `.takeaway-strip`. If you reach 4+ in a row, convert one slide to a plain visual or an unframed question.

# Five Slide Families

# Five slide families

| Family         | Goal                   | Typical component              |
|----------------|------------------------|--------------------------------|
| Interaction    | Prediction / MCQ / TPS | bold question, .panel-question |
| Data reveal    | New data or output     | full-width graph, plain slide  |
| Interpretation | Read the result        | .panel-result, .takeaway-strip |
| Rule /         | Heuristic,             | .card-                         |

## VARIETY OF FAMILIES

In one lecture, cycle through all five families. No block of 5+ slides should belong to only one family.

# Tools Overview

# `.box-*` aliases — when to use what

| Alias                      | Maps to                      | Weight          | Colour                      | When to use                       |
|----------------------------|------------------------------|-----------------|-----------------------------|-----------------------------------|
| <code>.box-question</code> | <code>.panel-question</code> | panel (lighter) | orange left border          | MCQ, reflection question          |
| <code>.box-result</code>   | <code>.panel-result</code>   | panel (lighter) | indigo left border          | main conclusion, analysis result  |
| <code>.box-data</code>     | <code>.panel-data</code>     | panel (lighter) | olive left border           | dataset description, data context |
| <code>.box-rule</code>     | <code>.card-rule</code>      | card (heavier)  | amethyst top strip + shadow | rule, heuristic                   |
| <code>.box-warning</code>  | <code>.card-warning</code>   | card (heavier)  | red top strip + shadow      | methodological warning            |



# .box-question — interaction, MCQ

## QUESTION FOR REFLECTION

What percentage of penguin body-mass variability is explained by bill length?

**A)** less than 30 %   **B)** 30–60 %   **C)** more than 60 %

```
1 ::: {.box-question}
2 [Question for reflection]{.panel-title}
3
4 Question text + choices A / B / C.
5 :::
```

= .panel-question · orange left  
border · heading {.panel-title}

# .box-result — main result

## WHAT WE SEE

Each millimetre of bill length corresponds on average to **+8.7 g** body mass (95% CI: 6.2–11.2 g). The model explains 45 % of body-mass variability.

```
1 ::: {.box-result}
2 [What we see]{.panel-title}
3
4 Main conclusion or analysis result.
5 :::
```

= .panel-result · indigo left border  
· heading {.panel-title} or  
{.panel-lead}

# .box-data — dataset description

## PALMER PENGUINS

344 penguins from three islands · 3 species

(*Adelie*, *Chinstrap*, *Gentoo*)

Variables: bill length and depth (mm), flipper length (mm), body mass (g), sex, year

```
1 ::: {.box-data}
2 [Dataset name]{.panel-title}
3
4 Brief description: number of
5 observations, variables, source.
6 :::
```

= .panel-data · olive left border ·  
heading {.panel-title}

# .box-warning — methodological warning

## CAUTION — CORRELATION ≠ CAUSATION

A longer bill does not imply that *the bill causes* greater body mass — both traits may reflect overall body size.

```
1 ::: {.box-warning}
2 [Warning heading]{.card-title}
3
4 Methodological error, misinterpretation,
5 data limitation.
6 :::
```

= .card-warning · **card** · red top  
strip + shadow · heading {**.card-**  
**title**}

# .box-rule — rule, heuristic

## MEAN OR MEDIAN?

If the data are symmetric → mean.

If the data are skewed or contain outliers → median.

```
1 ::: {.box-rule}
2 [Rule heading]{.card-title}
3
4 Condition → recommendation.
5 Or a verbal heuristic.
6 :::
```

= .card-rule · **card** · amethyst top  
strip + shadow · heading { .card-  
title }

# Combination: `.box-question` + `.takeaway-strip`

## QUESTION FOR REFLECTION

Why does bill length differ between penguin species and within a single species?

**Morphology as adaptation.** Bill length reflects life strategy — Gentoo penguins forage differently from Adelie.

```
1 ::: {.box-question}
2 [Question]{.panel-title}
3 Question...
4 :::
5
6 ::: {.takeaway-strip}
7 **Conclusion.** Closing the conclusion.
8 :::
```

`.takeaway-strip` after `.box-question` closes a slide sequence.  
One slide = one question + one conclusion.

# Fragment: incremental reveal of `.box-result`

Let's look at the linear model output.

## RESULT

Bill length coefficient: **+8.7 g/mm** (95% CI: 6.2–11.2 g). The effect is statistically and biologically significant.

```
1 Sentence before the pause.  
2  
3 ::: {.fragment}  
4 ::: {.box-result}  
5 [Result]{.panel-title}  
6 Text that is revealed.  
7 :::  
8 :::
```

`::: {.fragment}` wraps any `.box-*` block. Students see the sentence → click → result.

# Tool: plain question without a frame

**How does the confidence interval change if we double the sample size?**

**A) Narrows    B) Stays the same    C) Widens**

Plain slide: bold question + choices without a frame. Maximum visual contrast against slides with `.box-*` components.



# Roughnotation — use sparingly

Each millimetre of penguin bill length corresponds on average to +8.7 g body mass — the effect is biologically significant.

```
1 [text]{.rn-underline-indigo}  
2 [text]{.rn-box-orange}  
3 [text]{.rn-circle-amethyst}  
4 [text]{.rn-bracket-indigo}
```

```
1 [text]{.rn-highlight-orange}  
2 [text]{.rn-cross-graphite}
```

Colours: indigo, orange,  
amethyst, olive, graphite, red

**Sparingly:** max 1–2 effects per slide, 3–5 per lecture. Types: underline, box, circle, bracket, highlight, cross. Not as decoration — only for one key moment in a concept.

# Reference Slides: SCSS Components

# Cards

# Card: basic structure

## CARD HEADING

Card body — one or two short paragraphs. The font is intentionally smaller than the base slide size (0.82 em) to fit more content.

- Cards can also contain a list
- Each item on its own line

Syntax: `::: {.card-indigo} + [text]{.card-title}` for the coloured top strip — without the old `.slide-card` + variant combination.

# Card: .card-title

THIS IS .CARD-TITLE

`[text]{.card-title}` must be the **first thing** inside the card — it creates the coloured top strip.

```
1 ::: {.card-insight}
2 [Section name]{.card-title}
3
4 Card body.
5 :::
```

# Card: compact variant

## STANDARD

`.card-orange` — padding `0.7rem 1rem`.  
Suitable for longer content.

## COMPACT

`.card-orange .card-compact` —  
reduced padding (`0.45rem 0.75rem`).  
For denser layouts.

# Card: overview of variants

|                                                |                                              |                                                |                                          |                                            |                                        |
|------------------------------------------------|----------------------------------------------|------------------------------------------------|------------------------------------------|--------------------------------------------|----------------------------------------|
| <b>INDIGO</b><br><code>.card-indigo</code>     | <b>ORANGE</b><br><code>.card-orange</code>   | <b>AMETHYST</b><br><code>.card-amethyst</code> | <b>OLIVE</b><br><code>.card-olive</code> | <b>RED</b><br><code>.card-red</code>       |                                        |
| <b>QUESTION</b><br><code>.card-question</code> | <b>INSIGHT</b><br><code>.card-insight</code> | <b>WARNING</b><br><code>.card-warning</code>   | <b>DATA</b><br><code>.card-data</code>   | <b>RESULT</b><br><code>.card-result</code> | <b>RULE</b><br><code>.card-rule</code> |

Semantic variants are aliases: `.card-question` = orange, `.card-insight` = `.card-result` = indigo, `.card-warning` = red, `.card-data` = olive, `.card-rule` = amethyst. Shorthand: `.box-question`, `.box-result`, `.box-data`, `.box-warning`, `.box-rule`.

# Panels



# Panel: `.panel-result`

## KEY INSIGHT

Purple left border. For the main message of a block — conclusion, analysis result, or key number.

```
1 ::: {.panel-result}
2 [Heading]{.panel-title}
3 Panel text.
4 :::
```

`.panel-title` is `display: block;`  
gives the heading h3 appearance  
without `###`.

# Panel: `.panel-question`

## CONCLUSION OR OPTION

Orange left border. For conclusions, comparison results, or offering multiple answer choices.

```
1 ::: {.panel-question}
2 [Conclusion or option]{.panel-title}
3 Panel text.
4 :::
```

# Panel: `.panel-question` + `.panel-lead`

## Computation result

Large highlighted result — replaces the old `.answer-title`. Use at most once per slide.

```
1 ::: {.panel-question}
2 [Computation result]{.panel-lead}
3
4 Text below the result.
5 :::
```

`.panel-lead` is large and bold.  
Suitable for a key number or conclusion.

# Panel: `.panel-note` and `.panel-warning`

## INSTRUCTIONS OR OUTPUT

Neutral olive border. For instructions, workflow steps, or model output.

## CAUTION!

Red left border. For warnings, common errors, or deliberate alerts.

# Panel: overview of colour variants

## INDIGO

```
.panel-  
indigo
```

## ORANGE

```
.panel-  
orange
```

## AMETHYST

```
.panel-  
amethyst
```

## OLIVE

```
.panel-  
olive
```

## RED

```
.panel-red
```

Semantic variants are aliases: `.panel-result` = indigo, `.panel-question` = orange, `.panel-warning` = red, `.panel-note` = amethyst, `.panel-data` = olive. Shorthand (`.box-*`) wraps the matching panel.

# Takeaway strip

# Takeaway strip: standard

Sentence that leads into the conclusion — contextual paragraph above the strip.

**Main slide conclusion.** `.takeaway-strip` — indigo-to-orange gradient background, indigo left border. One per slide, placed at the bottom.

::: `{.takeaway-strip}` — the gradient is built in, no extra class needed.

# Takeaway strip: compact variant

Sentence above the strip as usual.

Main conclusion — standard strip.

Supplementary conclusion — `.strip-compact` reduces `margin-top`. Useful for two conclusions together.



# Strip: colour variants

`.strip-`  
`indigo`

`.strip-`  
`orange`

`.strip-`  
`amethyst`

`.strip-`  
`olive`

`.strip-red`

# Blockquote

# Blockquote div: `.blockquote`

**Definition or quotation.** `.blockquote` — light purple background, indigo left border (6px), rounded corners. Font 0.82 em.

- Can contain a list
- Each item on its own line

Use `::: {.blockquote}`, not Markdown `>`. The div variant gets correct styling; Markdown `blockquote` gets different styles from `presentation_theme.scss`.

# Slide backgrounds

# Backgrounds: dark variants

This slide has the class `{.bg-indigo}`. Text and headings are automatically white.

Syntax — add the class to the slide heading:

```
1 ## Slide title {.bg-indigo}
```

Available dark variants:

- `{.bg-indigo}` — primary indigo
- `{.bg-amethyst}` — lighter purple
- `{.bg-olive}` — grey-green
- `{.bg-red}` — red
- `{.bg-graphite}` — dark grey

# Backgrounds: light variants

This slide has the class `{.bg-parchment}`. Text is automatically graphite.

Available light variants:

- `{.bg-orange}` — orange
- `{.bg-parchment}` — cream
- `{.bg-white}` — pure white

Text and heading colour adapts automatically — no need to add `.text-color-*`.

# Inline highlights

## Inline highlight: `.text-hgl-{color}`

The model intercept is  $-153.4$  and the length coefficient is `4.05 g/mm` — each millimetre of length corresponds on average to four grams of body mass.

Caution — this procedure works `only when the normality assumption is met`.

Syntax: `[text]{.text-hgl-indigo_velvet}` — coloured chip with contrasting text.

Available colours: `indigo_velvet`, `orange`, `amethyst`, `grey_olive`, `graphite`, `white`, `light_gray`, `parchment`.



# Inline highlight: `.text-bg-{color}` and `.text-color-{color}`

`.text-hgl-indigo_velvet` — chip with contrasting text

`.text-bg-indigo_velvet` — chip without text colour change

`.text-color-indigo_velvet` — text colour only

`.text-color-indigo_velvet`  
`.text-bold` — colour + bold

```
1 [text]{.text-hgl-indigo_velvet}
2 [text]{.text-bg-indigo_velvet}
3 [text]{.text-color-indigo_velvet}
4 [text]{.text-color-indigo_velvet .text-bold}
```

# Roughnotation

# Roughnotation: shorthand syntax

Shorthand `{.rn-{type}-{colour}}` expands via `rn-shorthand.lua`:

## Short form (new):

Each millimetre of length corresponds to approximately four grams of body mass.

```
1 [text]{.rn-underline-indigo}
```

## Long form (still functional):

```
1 [text]{.rn-fragment rn-type=underline
2   rn-color="#5D2890"}
```

Available types: underline, box, circle, highlight, cross, bracket. Colours: indigo, orange, amethyst, olive, graphite, red.

# Roughnotation: box, circle, bracket

## box

$$R^2 = 0.76$$

```
1 [0.76]{.rn-box-indigo}
```

## circle

The relationship is  
strong.

```
1 [strong]{.rn-circle-orange}
```

## bracket

$$\hat{y} = \beta_0 + \beta_1 x$$

```
1 [\beta_1 x]{.rn-bracket-indigo}
```

Animation fires on fragment arrival. Do not use both colours on the same slide.

# Fragments and progressive disclosure

# Fragment: `.fragment` block

A sentence that is on the slide from the start.

## CONCLUSION ARRIVES AFTER PAUSE

This entire block appears as a single fragment — the student sees the sentence above and only after clicking does the panel with the conclusion reveal itself.

Syntax: `::: {.fragment}` wraps any block. Works with panels, blockquotes, strips, and cards.

# Fragment: `.incremental` list

Which properties must a good correlation coefficient satisfy?

- It must be symmetric: correlation of X with Y = correlation of Y with X.
- It must be dimensionless: independent of the scale of variables.
- It must be in the interval  $-1$  to  $+1$ .
- $+1$  = perfect positive dependence,  $-1$  = perfect negative dependence.

Each item appears on click.

# Typography



# Typography: sizes

`.text-size-heading` — 140 % of base

`.text-size-large` — 130 % of base

Normal text (100 %, base 30 px)

`.text-size-smaller` — 82 % of base

`.text-size-tiny` — 65 % of base

```
1 [text]{.text-size-heading}
2 [text]{.text-size-large}
3 [text]{.text-size-smaller}
4 [text]{.text-size-tiny}
```

`.text-size-body` = 100 % (explicit reset). Old `.text-larger` / `.text-smaller` / `.text-tiny` are removed.

# Typography: weight, alignment, font

`.text-center` — horizontal centre

`.text-right` — right alignment

`.text-font-heading` — Source Sans 3, heading weight

`.text-font-monospace` — JetBrains Mono

`.text-bold` — **bold**

# Colour utilities

# Colours: indigo\_velvet and amethyst

`.bg-indigo_velvet` — primary  
indigo

`.text-color-indigo_velvet`

`.text-hgl-indigo_velvet`

`.bg-amethyst` — lighter purple

`.text-color-amethyst`

`.text-hgl-amethyst`

# Colours: orange and graphite

`.bg-orange` — vivid orange

`.text-color-orange`

`.text-hgl-orange`

`.bg-graphite` — dark base

`.text-color-graphite`

`.text-hgl-graphite`

# Colours: neutral palette tones

`.bg-parchment`

`[.text-color-  
parchment]`

`.text-hgl-  
parchment`

`.bg-grey_olive`

`.text-color-  
grey_olive`

`[.text-hgl-  
grey_olive]`

`.bg-light_gray`

`[.text-color-  
light_gray]`

`.text-hgl-  
light_gray`

`.bg-white`

`[.text-color-  
white]`

`.text-hgl-white`

# Layout utilities

# Layout: `.columns.align-top`

**Left column** starts at the top — without `.align-top` the content would be vertically centred.

Card in the left column.  
Second paragraph inside the card — the left column is taller.

**Right column** is shorter.

Without `.align-top` the right text would shift down to the centre of the column pair.

Syntax: `::: {.columns .align-top}` — add the class `.align-top` to `.columns`.



# Layout: vertical offsets

Example: `.slide-margin-top-15`

This card would have `margin-top: 15%` if it had the class `.slide-margin-top-15`.

Useful for vertically centring an isolated element on a slide.

`.slide-margin-top-15` → `margin-top: 15%` · `.slide-margin-top-25` → `margin-top: 25%` · `.center-vertical` → absolute centre

For a group of related elements, prefer the normal-flow `.layout-stack.layout-center-y` utility shown below.

# Layout: `.layout-grid-2`

## LEFT RESULT

Two equal-width elements form one comparison.

## RIGHT QUESTION

The shared gap keeps the pair readable.

Use `::: {.layout-grid-2}` for a comparison or prompt pair—not merely to decorate an ordinary list.

# Layout: vertical stack

First related teaching element.

Second element with consistent breathing room.

The default gap is `1.4rem`; override `--layout-gap` on the wrapper only when the rendered composition needs it.

# Layout: centred stack

Which observation would change your conclusion?

`.layout-center-y` centres the short stack in normal flow without transforming its children.